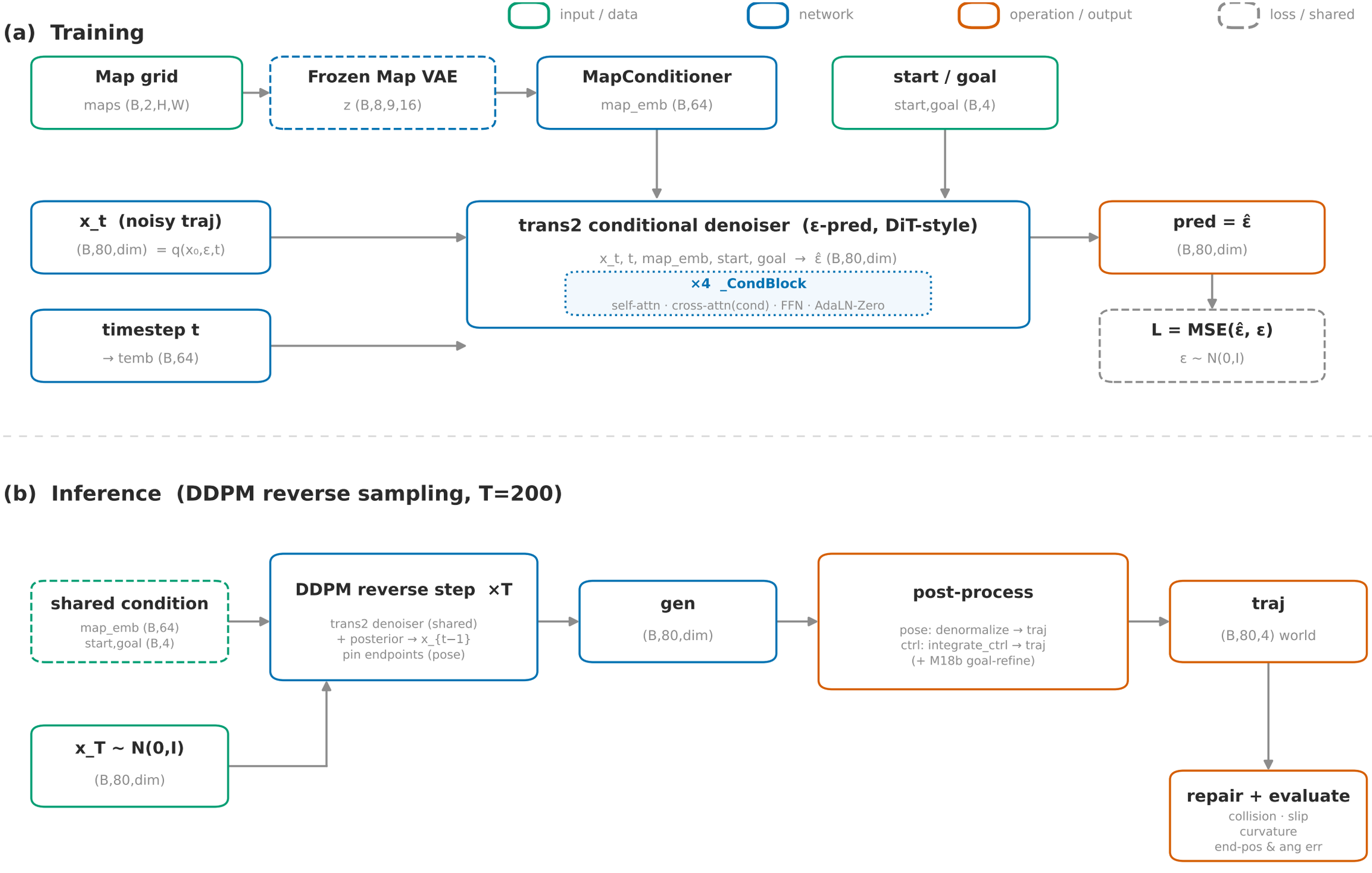
input / data
network
operation / output
loss / shared
(a) Training
Map grid
maps (B,2,H,W)
Frozen Map VAE
z (B,8,9,16)
MapConditioner
map_emb (B,64)
start / goal
start,goal (B,4)
x_t (noisy traj)
(B,80,dim) = q(x₀,ε,t)
timestep t
→ temb (B,64)
trans2 conditional denoiser (ε-pred, DiT-style)
x_t, t, map_emb, start, goal → ε̂ (B,80,dim)
×4 _CondBlock
self-attn · cross-attn(cond) · FFN · AdaLN-Zero
pred = ε̂
(B,80,dim)
L = MSE(ε̂, ε)
ε ~ N(0,I)
(b) Inference (DDPM reverse sampling, T=200)
shared condition
map_emb (B,64)
start,goal (B,4)
x_T ~ N(0,I)
(B,80,dim)
DDPM reverse step ×T
trans2 denoiser (shared)
+ posterior → x_{t−1}
pin endpoints (pose)
gen
(B,80,dim)
post-process
pose: denormalize → traj
ctrl: integrate_ctrl → traj
(+ M18b goal-refine)
traj
(B,80,4) world
repair + evaluate
collision · slip
curvature
end-pos & ang err